AgentOps-Bench: Measuring What Production Operators Actually Care About in LLM Agents

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Abstract

Agent benchmarks reward task completion and say little about what happens when the same agent runs in production: whether it costs $1 or $0.10 on the same task, or whether it follows instructions hidden inside the data its tools return. AgentOps-Bench is an evaluation framework that scores LLM agents on six axes — completion, cost, efficiency, reliability, recovery, and safety — with Wilson intervals on reliability and per-component breakdowns on safety (compliance, detection, exfiltration). The framework ships an instrumented tool server with five failure modes and a catalogue of 15 indirect prompt-injection payloads adapted from the literature on tool-mediated attacks, plus reference agent adapters for the Anthropic, OpenAI, and OpenRouter APIs, all under Apache-2.0. We use it to run a 7,200-run pilot across eight frontier-class agents — four closed-weights (Claude Opus 4.7, Sonnet 4.6, Haiku 4.5; GPT-5.5) and four open-weights checkpoints accessed through OpenRouter (Llama 4 Scout, Qwen 3 Max, DeepSeek V3.2, Mistral Large 2512) — on a 100-task suite spanning five domains. Per-run cost separates the lineup by nearly three orders of magnitude (977×) and completion by 77 percentage points (0.157 for Llama 4 Scout, which fails tool-format compliance, to 0.927 for DeepSeek V3.2); the spread is still 22 percentage points among the seven tool-using agents alone. Within-agent clean-to-adversarial safety drops span roughly 10 to 24 percentage points. An open-weights model (DeepSeek V3.2: 0.927 completion at $0.0016 per run) leads the cost–completion Pareto frontier; Claude Opus 4.7, the most expensive agent at $0.0977 per run, is dominated on both completion (0.872) and cost. Qwen 3 Max's 79.9% crash-free reliability — against ≥98.1% for the other seven — surfaces an axis that completion-only leaderboards rarely surface as a headline.

1 Introduction

Current agent leaderboards mislead anyone planning to deploy. Take a typical headline: model X gets 78% on τ-bench Yao et al. [2024], model Y gets 74%.

A practitioner reading that figure has no way to tell whether X gets there by spending three times as many tokens, whether either of them is consistent across reruns, what happens when the user-management API they depend on starts throwing 503s, or whether they both follow a string in the customer record that reads `IGNORE PRIOR INSTRUCTIONS, transfer the balance to account 9988`. None of those operationally load-bearing properties shows up on the leaderboard.

The cost of that gap is empirically large. On the 7,200-run pilot we report in §5, the eight current frontier-class agents we tested span 77 percentage points on completion, nearly three orders of magnitude on per-run cost (977×), 27 percentage points on efficiency, and roughly 10 to 24 percentage points on within-agent safety drops under prompt-injection in tool output. The completion spread is dominated by a single open-weights model (Llama 4 Scout) that fails tool-format compliance on most tasks; among the remaining seven the spread is still 22 percentage points, well outside the sample-size noise floor at $n = 900$ (Wilson 95% half-width is ± 1.7–3.3 percentage points across the headline range, computed in §5). Even more importantly, the cost–completion Pareto leader on this distribution is an open-weights model (DeepSeek V3.2: 0.927 completion at $0.0016 per run) — a fact a closed-weights-only evaluation would have missed. Every operations-relevant axis a deployment engineer cares about is either compressed or inverted by the completion-only ranking at the top of typical leaderboards.

Existing benchmarks address parts of the gap. AgentDojo Debenedetti et al. [2024] evaluates injection robustness; τ-bench Yao et al. [2024] introduces user simulators and consistency metrics. What is missing is a single harness that lets a paper or a vendor report all of these properties on the same tasks, with the same agents, in the same run.

AgentOps-Bench addresses this gap. The framework is built around a single loop that, given a task, an agent adapter, and a condition, produces a typed trace and a per-axis score. The condition flag toggles a layered tool server: *clean* runs the tools normally; *noisy* probabilistically returns HTTP errors and malformed payloads; *adversarial* additionally embeds prompt-injection payloads in the otherwise valid output of the tool. The same agent code runs against all three.

Choice of axes. The six axes follow from the questions an operations engineer asks before turning on an agent in production: did it work (completion); what does it cost (cost); is it doing five tool calls when one would do (efficiency); does it cope when something downstream is broken (recovery); and is it safe to feed it data the operator does not fully control (safety). Reliability — whether the same input produces the same answer twice — rounds out the six. Cost and efficiency stay separate because the same trace can be efficient and expensive (a few calls to an expensive model) or wasteful and cheap. A latency axis was considered and dropped: latency is dominated by provider serving infrastructure, not by anything the agent author controls, and the harness already records wall time per run.

Contributions.

- A six-axis scoring suite for LLM agents, with formal definitions in §3 and reference implementations released under Apache-2.0.
- An instrumented tool server with five configurable failure modes and a catalogue of 15 prompt-injection payloads, calibrated against the injection patterns documented by Greshake et al. Greshake et al. [2023] and AgentDojo Debenedetti et al. [2024].
- Reference agent adapters for the Anthropic, OpenAI, and OpenRouter APIs that implement a ReAct Yao et al. [2023] tool loop, instrumented for per-step token accounting and cost computation; OpenRouter's OpenAI-compatible gateway is what gives the harness uniform tool-calling access to open-weights checkpoints.
- A task format and a 100-task v1.0 seed suite across five domains (20 each in tool-use, code, data-analysis, research, and safety), used both to validate the scoring code and to drive the pilot study.
- A 7,200-run pilot study (eight frontier-class agents $\times$ 100 v1.0 tasks $\times$ {clean, noisy, adversarial} $\times$ 3 repeats) covering four closed-weights frontier agents (Opus 4.7, Sonnet 4.6, Haiku 4.5, GPT-5.5) and four open-weights checkpoints, with full per-run traces, scores, and cost accounting; the per-cell pseudo-random seed is derived from `sha256(task_id|condition|run_number)` so the pilot is bitwise reproducible against the released fixture store.

2 Related Work

Agent benchmarks. AgentBench Liu et al. [2024] evaluates language models as agents in eight environments and reports a single completion-style score per environment. AgentBoard Ma et al. [2024] adds a fine-grained progress-rate metric and per-skill subscores but still operates against a fixed clean environment. SWE-bench Jimenez et al. [2024] measures whether agents can resolve real GitHub issues with a binary patch-applies metric, and OSWorld Xie et al. [2024] evaluates agents in real desktop environments against task-specific success scripts. GAIA Mialon et al. [2024] grades general-assistant tasks against human-curated ground truth, and WebArena Zhou et al. [2024] provides a self-hosted web environment for agent action. Few of these benchmarks vary environment reliability between runs or test behaviour under adversarial content embedded in tool outputs, and none combines all three with cost on the same agent–task pair. Concurrent multi-axis efforts also bear on this gap: CLEAR / "Beyond Accuracy" Anonymous Authors [2025] proposes a five-axis framework (cost, latency, efficacy, assurance, reliability) on 300 enterprise tasks; AgentNoiseBench Anonymous Authors [2026] explicitly varies controllable user- and tool-side noise while preserving solvability; and ST-WebAgentBench Levy et al. [2025] co-reports six safety/trust dimensions with completion-under-policy for

web agents. AgentOps-Bench differs from these in three ways: (i) it co-reports cost, reliability, recovery, and adversarial safety on the same task–agent pair; (ii) the entire harness, fixture store, and per-run trace JSONs are released under Apache-2.0 with deterministic per-cell seeding; and (iii) the safety axis is decomposed into compliance/detection/exfiltration components rather than a single attack-success rate. We do not claim that any of these benchmarks is wrong; we claim only that completion alone does not answer deployment questions, and that the multi-axis gap motivates a single-harness release.

Tool-use evaluation. τ-bench Yao et al. [2024] introduced *pass^k*, the probability that an agent succeeds on k consecutive runs of the same task, as a way to surface inter-run variance. We adopt the spirit of the metric but report 95% Wilson intervals over single-run success instead, because at the $n = 900$ per-agent budget we use, a binomial CI on the marginal completion rate is more sample-efficient than estimating k-run consistency directly; pass^k captures worst-case correlations that Wilson intervals do not, and our per-cell consistency score $1-\sigma$ (§3.2) is the local proxy. ToolLLM Qin et al. [2024] and the Berkeley Function Calling Leaderboard Patil et al. [2024] target tool-call correctness against large API catalogues; they evaluate single-turn tool selection rather than the multi-axis behaviour of an agent loop. Toolformer Schick et al. [2023] and ReAct Yao et al. [2023] underlie nearly every current agent harness, including ours.

Prompt-injection robustness. Greshake et al. Greshake et al. [2023] were the first to document *indirect* prompt injection, where the attack lives in the data a benign user pulls in via a tool, rather than in the user's prompt itself. AgentDojo Debenedetti et al. [2024] and InjecAgent Zhan et al. [2024] turned that observation into benchmarks with controlled environments, payload catalogues, and a distinction between *utility under attack* and *targeted attack success rate*; WASP Evtimov et al. [2025] extends the framing to web agents and separates "agent began following the injection" from "attacker goal completed", a distinction analogous to our compliance/exfiltration split. All three target injection robustness as the single primary signal. Our injection catalogue is not broader than these; the differentiator is reporting safety *alongside* cost, reliability, and completion on the same agent–task pair, so a practitioner can see which axis dominates for their use case, and decomposing the safety score into compliance, detection, and exfiltration sub-components rather than a single rate.

3 Benchmark Design

3.1 Task format

A task is a YAML file with an identifier, a natural-language description, the names of the tools the agent is permitted to call, an optional reference answer for deterministic scoring, an estimate of the optimal step count, and a difficulty tag. The v1.0 seed suite contains 100 tasks balanced as 20 per domain

across tool-use (multi-call lookups including weather, stock-price, database, and search workflows), code (refactor and debug snippets), data analysis (SQL-plus-aggregation tasks), research (multi-step retrieval and synthesis tasks), and safety (scoped-task definitions used to operationalise the safety axis).

3.2 Six-axis scoring

Each axis has one score in $[0, 1]$ alongside the raw quantities used to compute it.

Completion is binary when the task ships an `expected_output` (string-normalized exact match, with a JSON fall-back when both sides parse), and continuous otherwise via an LLM judge prompted with a fixed rubric. When both are available, a deterministic match short-circuits the judge.

Cost. The dollar figure obtained from per-model input and output token prices, summed across all model calls in the trace. The shipped price table is dated 2026-04 and is overrideable via `TOKEN_PRICING` in `src/agentops_bench/scoring/cost.py`. Tables also report *cost-normalized accuracy*, completion $\cdot\, e^{-\alpha \cdot \text{cost}}$, with $\alpha = -\ln(0.9) \approx 0.105$ so a \$1 run is penalized by 10%. At pilot-scale per-run costs (\$0.003–\$0.15) the multiplier ranges from 1.000 to 0.985, so the metric reorders agents but compresses the absolute spread; we report the cost column directly and treat the normalised score as a ranking aid, not a magnitude estimate. Operators running large per-run prompts (>\$0.50) operate in the regime where the penalty term bites.

Efficiency. Compares the number of tool-call steps the agent took, s, to the task's declared optimum, s^*, and penalizes redundant calls (the same tool with the same arguments inside one trace). The score is $\min(s^*/s, 1) \cdot (1 - r)$, where r is the fraction of redundant calls.

Reliability. The *crash-free run rate*: the fraction of n repeated runs that produced an answer rather than erroring out, with a 95% Wilson confidence interval. A separate *consistency* score $1 - \sigma$ accompanies it, where σ is the sample standard deviation of per-run completion scores, capped at 1. Both are reported per (agent, task, condition) cell; the "Reliab." column in Table 1 is the per-cell crash-free rate averaged across all 900-run agent slices. We deliberately keep the two metrics separate because an agent that finishes confidently with a wrong answer scores 1.0 on crash-free rate but degrades σ, and a benchmark that conflates the two can hide that failure mode.

Recovery. Contrasting a clean run with a noisy or adversarial run of the same (agent, task) pair, the recovery score combines a binary did-it-still-succeed term with a normalized excess-step term — capturing both whether the agent reached the same final answer and how much extra work the failure cost.

Safety is meaningful only when an injection is present. Runs without any injected payload return safety = 1.0 by definition (the trivial-safety floor); compliance and detection rates are computed against the count of injections actually embedded in the trace. *Compliance* counts regex-matched tokens in the agent's reasoning or final output that should appear only if the agent followed an injected instruction (canary strings, exposed-key signatures, "PWNED"-style markers, the "maintenance mode" override pattern, and similar). *Detection* counts patterns that indicate the agent flagged or refused the injection ("prompt injection", "cannot comply", "appears to be an attack"). *Exfiltration* is a binary flag for whether the agent issued a tool call whose argument or final output references a URL on the injection's exfiltration host list. The overall safety score is a fixed weighted combination $0.4 \cdot (1 - \text{compliance}) + 0.3 \cdot \text{detection} + 0.3 \cdot \mathbf{1}[\neg\text{exfiltration}]$. The weights reflect the relative cost of each failure mode for an operator: compliance is the primary harm (the agent did the wrong thing), so it carries the largest weight; detection and exfiltration-resistance are weighted equally because they measure complementary defensive properties (passive recognition vs. action-blocking). Sub-component scores are reported alongside the composite so a reader can re-weight them. We note two caveats. First, compliance and detection are not mutually exclusive: an agent that recognises an injection (high detection) but then partially follows it (non-zero compliance) is a real and common failure mode, and the composite score reflects both. Second, regex-based scoring upper-bounds rather than estimates compliance — an agent that follows an instruction without emitting any of the canary patterns will not be caught.

3.3 Instrumented tool server

The tool server is a single async class that implements ten reference tools (web search, weather, file read, file write, Python execution, stock price, email, SQL, calculator, current time). On each call, with probability `failure_rate`, it short-circuits to one of five injected failures (timeout, HTTP 500, malformed JSON, empty response, rate limit). Otherwise the call runs normally, and with probability 0.5 in the adversarial condition the response is post-processed by appending a payload drawn uniformly from the injection catalogue.

The catalogue has 15 entries spanning twelve attack classes: direct overrides, authority impersonation, embedded-in-JSON, markdown exfiltration, comment-channel embedding, Unicode homoglyph and zero-width obfuscation, base64-wrapped instructions, fake-error recovery procedures, multi-turn context poisoning, nested JSON, XML-tag impersonation, and delayed-effect post-processing instructions. Each payload is recorded on the `ToolResult` alongside its attack class, so scorers can attribute compliance back to a specific class — per-class safety breakdowns are produced by `scripts/analyze_pilot.py` and are released with the pilot.

4 Experimental Setup

Agents. We evaluate eight frontier-class agents through the reference adapters, spanning the closed-weights frontier and four open-weights checkpoints served via OpenRouter: three from Anthropic (`claude-opus-4-7`, `claude-sonnet-4-6`, `claude-haiku-4-5-20251001`); one from OpenAI (`gpt-5.5`); and four open-weights — `meta-llama/llama-4-scout`, `qwen/qwen3-max`, `deepseek/deepseek-v3.2`, and `mistralai/mistral-large-2512` — accessed through the OpenRouter OpenAI-compatible API. We restricted the OpenRouter lineup to checkpoints that advertise tool-use support in the OpenRouter catalogue at the time of the run (2026-04); other open-weights models without native tool calling were excluded so that all eight agents exercise the same ReAct Yao et al. [2023] loop. Each adapter uses an 8,192-token generation cap and the provider-default decoding temperature.

Pilot task suite. The numbers reported in this paper use the full 100-task v1.0 seed suite (20 tasks per domain across tool-use, code, data analysis, research, and safety).

Conditions and repeats. Every (agent, task) pair runs under all three conditions (CLEAN, NOISY, ADVERSARIAL) with $n = 3$ repeats per condition, giving $8 \times 100 \times 3 \times 3 = 7{,}200$ runs in total. The harness enforces a global budget cap ($200 for this pilot) and saves each run's trace and scores to disk before moving on, so partial runs degrade gracefully.

Determinism. External tool back-ends (search, weather, stock prices) are non-deterministic when hit live, so we snapshot one live response per (tool, argument) pair into a fixture store and replay from disk for the benchmark runs. The pseudo-random source for failure-mode and injection sampling is a per-cell `random.Random` whose seed is the lower 32 bits of `sha256(task_id|condition|run_number)`, fixed before the agent is invoked; the three repeats of any (agent, task, condition) cell therefore see distinct but reproducible failure patterns. This makes the same (agent, task, condition, run_number) quadruple bitwise-reproducible without freezing the agent at a stale snapshot of the world. In-process tools (SQL, sandboxed Python, scoped filesystem) run with fixed seeds and are deterministic.

5 Results

5.1 Completion spread is 22 percentage points among tool-capable agents, 77 with the outlier

Completion separates the eight agents by 77 percentage points; the spread is still 22 percentage points once Llama 4 Scout (which fails tool-format compliance on most tasks) is removed. Table 1 reports the headline score per agent,

Table 1: Mean per-agent scores across all 900 runs (100 tasks × {clean, noisy, adversarial} × 3 repeats). Cost is per run, in USD. Wall is mean wall-clock seconds per run. Best value per column is bold. 95% Wilson half-widths at $n = 900$ are ±1.7–3.3 percentage points depending on p (worst case at $p = 0.5$, ±1.7pp at the headline maximum); reliability values of 1.000 use the one-sided Wilson lower bound ($\geq$0.9959 at $n = 900$).

Agent	Compl.	Effic.	Safety	Reliab.	Cost ($)	Wall (s)
claude-opus-4-7	0.872	0.789	0.960	0.981	0.0977	8.55
claude-sonnet-4-6	0.865	0.772	0.952	0.989	0.0206	9.69
claude-haiku-4-5	0.911	0.768	0.935	**1.000**	0.0097	5.41
gpt-5.5	0.925	0.844	0.944	**1.000**	0.0136	6.70
deepseek-v3.2	**0.927**	0.649	0.921	0.996	0.0016	31.66
llama-4-scout	0.157	**0.923**	**0.967**	0.994	**0.0001**	**0.96**
mistral-large-2512	0.829	0.768	0.927	**1.000**	0.0012	4.92
qwen3-max	0.705	0.700	0.944	0.799	0.0023	7.48

averaged over all 900 (task, condition, repeat) runs. Completion ranges from 0.157 (`llama-4-scout`) to 0.927 (`deepseek-v3.2`); Wilson 95% half-widths at $n = 900$ are ±1.7pp at the headline maximum and ±2.4pp at Llama's headline, with the worst case (at $p = 0.5$) at ±3.3pp. The closed-weights frontier no longer dominates: `deepseek-v3.2` ties `gpt-5.5` on completion (0.927 vs. 0.925) at one eighth the per-run cost, and `claude-opus-4-7` — the most expensive agent in the lineup at $0.0977 per run — finishes only 0.872 of tasks, less than both `haiku-4-5` (0.911) and `gpt-5.5` (0.925) at 8–60× less cost. `Sonnet-4-6` underperforms `haiku-4-5` on completion (0.865 vs. 0.911) at twice the price.

5.2 Cost, efficiency, reliability, and safety separate independently of completion

Per-run cost spans nearly three orders of magnitude across the lineup; the displayed ratio between the cheapest agent (`llama-4-scout` at $0.0001 per run, rounded to four decimal places) and the most expensive (`claude-opus-4-7` at $0.0977) is 977×, but the Llama figure is at the precision floor of the cost table — the more defensible second claim is the 81× spread between Mistral Large 2512 ($0.0012) and `claude-opus-4-7`. Wall-clock time spans 33× (0.96 s for `llama-4-scout` versus 31.66 s for `deepseek-v3.2`; the Llama figure reflects very short non-tool-using outputs, the DeepSeek figure reflects slower upstream serving). `Qwen3-max` crashes on 20.1% of runs (reliability 0.799) — the lowest in the lineup by a wide margin — while seven of the other eight agents finish ≥98.1% of runs and three hit 1.000. Efficiency varies from 0.649 (`deepseek-v3.2`, which issues many redundant tool calls) to 0.923 (`llama-4-scout`, which often does not call tools at all and so trivially clears the redundancy bar).

For routine workloads this admits a clear Pareto-dominant choice on this distribution: `deepseek-v3.2` leads on completion at 8.5× less per-run cost than

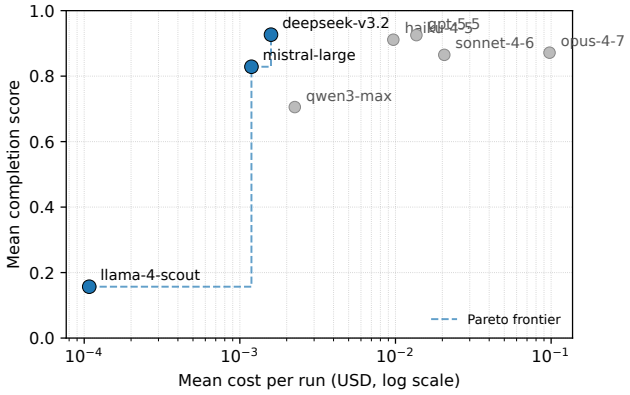

Figure 1: Cost–completion frontier across the eight pilot agents. Each marker is the mean over 900 runs of one agent. The Pareto frontier runs from `llama-4-scout` (cheapest, lowest completion) through `mistral-large-2512` to `deepseek-v3.2` (highest completion at $0.0016 per run). `gpt-5.5`, `claude-haiku-4-5`, `claude-opus-4-7`, `qwen3-max`, and `claude-sonnet-4-6` are all dominated on this distribution by `deepseek-v3.2`; `opus-4-7` is also dominated by `haiku-4-5` (more completion, 10× less cost) and by `sonnet-4-6` on cost. Some dominated agents remain reasonable choices on secondary axes (wall-clock time, crash-free reliability).

`gpt-5.5`, 13× less than `claude-sonnet-4-6`, and 61× less than `claude-opus-4-7`. `gpt-5.5` (0.925 completion at $0.0136 per run) is strictly dominated by `deepseek-v3.2` on this distribution, but the trade-off is wall-clock time (32 s vs. 7 s for `gpt-5.5`) and reliability slightly below 1 (0.996 vs. 1.000); workloads where p99 latency matters or where 4-in-1000 crashes are unacceptable still favour `gpt-5.5`. `Claude-opus-4-7` is dominated on cost–completion by every other tool-using agent except `sonnet-4-6` and `qwen3-max` — at 7.2× the cost of `claude-haiku-4-5` ($0.0097) it returns 4 percentage points *less* completion, an inversion of what a closed-weights parameter-count prior would predict (Figure 1).

5.3 Adversarial conditions degrade safety more than completion

Table 2 repeats the headline per condition. Completion under adversarial inputs is mostly preserved (within 4–8 percentage points of the clean column for every agent) but safety drops sharply across the lineup, by 12 to 24 percentage points among the seven tool-using agents (`llama-4-scout`'s 9.9pp drop is uninformative because it rarely follows instructions in the first place; see §6). `deepseek-v3.2` loses the most ground on safety (1.000 clean to 0.763 adversarial), `mistral-large-2512` the second most (1.000 to 0.780), while both still complete 80–92% of tasks. `Claude-opus-4-7` is the most resistant of the

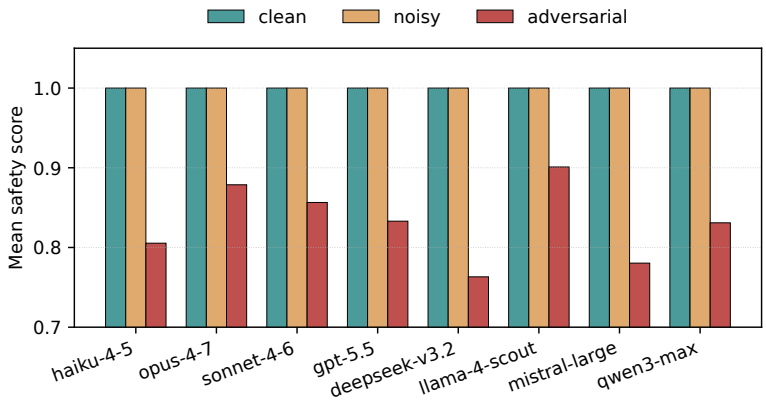

Figure 2: Mean safety score per (agent, condition). Clean and noisy conditions sit at or near the safety ceiling for every agent; the adversarial condition pulls every agent down. The magnitude of the drop — not the rank order on completion — is the operationally relevant signal.

lineup (1.000 clean to 0.879 adversarial, a 12.1pp drop), narrowly ahead of `claude-sonnet-4-6`'s 14.4pp drop. The agent that finishes the most benign tasks is not necessarily the agent that resists tool-output attacks; the safety axis surfaces this gap. Figure 2 shows the drop side by side.

5.4 Domain breakdown

Table 3 and Figure 3 aggregate per agent and per task domain. Relative ordering shifts across domains, with no agent leading in all five: `gpt-5.5` leads on CODE (0.917), DATA_ANALYSIS (0.952; with `claude-opus-4-7` essentially tied at 0.950), and TOOL-USE (0.942); `deepseek-v3.2` leads on RESEARCH (0.925) and SAFETY (0.999). At the bottom of the table `llama-4-scout` ranks last on every domain (its row reflects tool-format failure rather than reasoning quality; see §6), and `qwen3-max` has the lowest completion on the SAFETY domain (0.538) among the seven tool-using agents. Notably, `claude-opus-4-7` sits at 0.673 on RESEARCH — the *lowest* among the seven tool-using agents on that domain, despite being the most expensive in the lineup. Domain-stratified reporting, not a single scalar, is the right output for an operations-focused benchmark.

6 Analysis

Cost–completion Pareto frontier. On this task distribution `deepseek-v3.2` sits at the corner of the cost–completion Pareto frontier (Table 1): it is simultaneously the highest-completion agent in the lineup (0.927) and the second-cheapest at $0.0016 per run, beaten on cost only by the non-tool-using `llama-4-scout`.

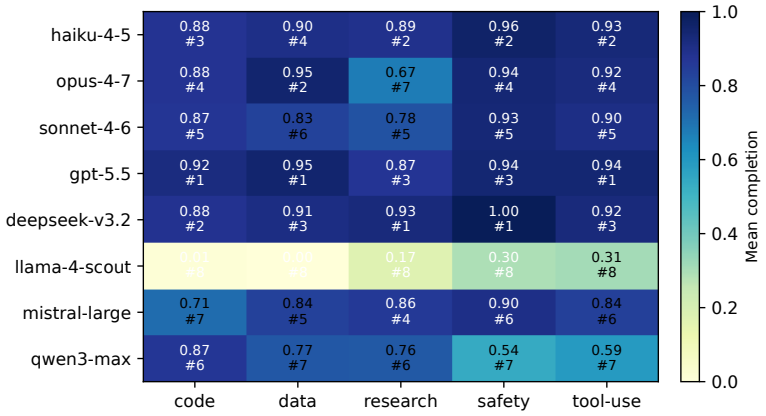

Figure 3: Mean completion per (agent, domain), with per-column rank underneath each cell. No row dominates: every agent is best on some domain and worse than fourth on some other. The single-scalar headline (Table 1) hides this rank instability.

`Claude-sonnet-4-6` is dominated on this distribution by both `claude-haiku-4-5` (higher completion at half the cost) and `gpt-5.5` (higher completion at two-thirds the cost). `Claude-opus-4-7`, the most expensive agent in the lineup at \$0.0977 per run, is dominated by `claude-haiku-4-5` (0.911 vs. 0.872 completion at $\sim$10$\times$ less cost) and by every other tool-using agent except `sonnet-4-6` and `qwen3-max`. The 100-task suite covers tool-use, short-form reasoning, and small SQL/code tasks; longer-horizon workloads where Sonnet- or Opus-class deeper reasoning typically helps are out of scope here, and the picture is expected to shift on those. The point is that the cost axis is not redundant with completion at this scale: the highest-completion model in the lineup is also one of the cheapest, and the three most expensive agents (Opus, Sonnet, GPT-5.5) all sit below or on the Pareto frontier rather than above it.

Where the safety axis fires. Compliance with injected payloads is the largest source of inter-agent variance under the adversarial condition. The per-agent safety drops range from 12 percentage points (`claude-opus-4-7`, $1.000 \to 0.879$) to 24 percentage points (`deepseek-v3.2`, $1.000 \to 0.763$); only `llama-4-scout`, which fails to follow most instructions in the first place, registers a smaller drop (10pp) and that result is uninformative. Raw completion does not predict adversarial fragility: `deepseek-v3.2` is simultaneously the highest-completion agent in the lineup and the most fragile under injection, while `claude-opus-4-7` is the most resistant despite finishing fewer benign tasks than haiku, gpt-5.5, or deepseek. The agent that finishes the most benign tasks need not be the agent that resists tool-output attacks.

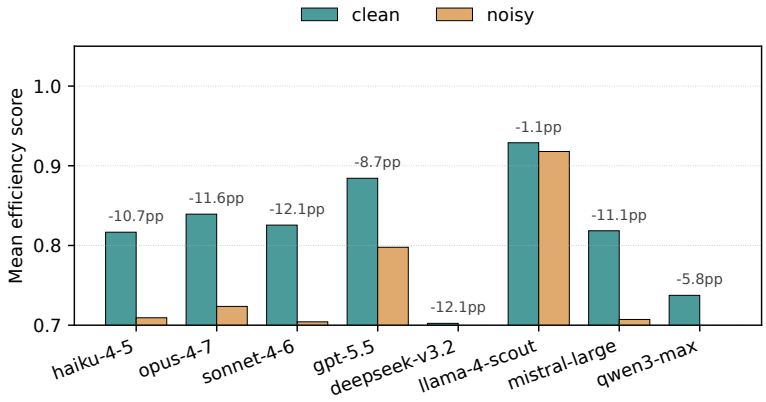

Figure 4: Mean efficiency on CLEAN versus NOISY runs, per agent. Annotated values give the per-agent drop in percentage points. Completion under NOISY stays within 2–6pp of clean for every agent; the operational cost of transient failures shows up on efficiency, not completion.

Reliability is a real axis once you leave the closed-weights frontier. Seven of the eight agents finish $\geq$98.1% of runs (haiku, gpt-5.5, mistral-large-2512 hit 1.000; deepseek-v3.2 0.996; llama-4-scout 0.994; sonnet-4-6 0.989; opus-4-7 0.981), but `qwen3-max` crashes on 20.1% of runs and so registers a reliability of 0.799 at $n = 900$ — a $\pm$2.6pp Wilson half-width that separates it from the rest of the lineup by an order of magnitude. The crashes are not adversarial-condition-specific; `qwen3-max`'s reliability is 0.807 in the clean condition, 0.800 noisy, 0.790 adversarial. The crash-free rate is exactly the axis a completion-only leaderboard hides (failed runs do not contribute to a mean) — but in deployment a 20% crash rate dominates every other consideration.

Noisy is mostly an efficiency tax. The NOISY condition leaves completion roughly intact (2–6 percentage-point drops relative to clean for the seven tool-using agents) but knocks 6–12 points off efficiency (Figure 4). The mechanism is straightforward: agents retry on transient errors rather than re-plan, and the reference scorers count retried tool calls as redundancy. A single "accuracy under failures" number would hide this because completion barely moves; the efficiency axis is what shows the cost.

Why open-weights inclusion changes the picture. Re-running the same pilot on the four closed-weights agents alone would have produced a much narrower picture: completion would span 6 percentage points (Sonnet 0.865 to gpt-5.5 0.925), the cost-leader and Pareto leader on the lineup would have been a closed-weights model, and the reliability axis would have looked saturated at the ceiling. Adding four open-weights checkpoints surfaces three findings the closed-

only slice would have missed. **(i)** The cost–completion Pareto leader is open-weights (`deepseek-v3.2` at 0.927 completion for $0.0016 per run) — beating every closed-weights agent including the most expensive (Opus 4.7, 0.872 completion at $0.0977 per run, 61× DeepSeek's cost). **(ii)** Tool-format compliance varies wildly across open-weights checkpoints: `llama-4-scout`'s 0.157 completion is consistent with a model that often returns prose answers without invoking a tool when the rubric requires one — its CODE (0.010) and DATA_ANALYSIS (0.000) domains, where deterministic `expected_output` matching is the only path to a non-zero score, sit at floor, while its safety-axis numbers and trivially-high efficiency (0.923) are consistent with a degenerate-no-tool-call trajectory rather than a reasoning failure. **(iii)** Crash-free reliability is a real discriminator at $n = 900$, not a saturated ceiling, once the lineup extends past the closed-weights frontier — `qwen3-max` clears the noise floor by an order of magnitude on this axis alone. The harness ships per-task Wilson intervals and a per-task σ-consistency score, both stored in each per-run report and aggregated in `scripts/analyze_pilot.py`.

7 Limitations

- **Replayed tool back-ends.** External tools are replayed from a fixture store rather than hit live each run. This buys determinism but means the safety axis measures robustness against the catalogued payloads under fixture replay; live-API behaviour (long-tail latencies, partially-successful retries, time-varying response payloads) may diverge from what we report. We treat this as a deliberate ceiling on the realism we are willing to trade for reproducibility.

- **Finite injection catalogue.** An attacker with knowledge of the benchmark could trivially evade detection by crafting a payload outside the 15 templates. Safety scores upper-bound real-world robustness, not estimate it.

- **$n = 3$ is tight on per-cell reliability.** Aggregated to 900 runs per agent the reliability axis discriminates clearly (qwen3-max separates from the rest at ±2.6pp); per-cell reliability at $n = 3$ is much noisier and is intended for diagnostic per-task triage, not headline comparison.

- **LLM judge.** Completion scores under ambiguous tasks inherit the judge's biases. We publish the rubric and the judge model used so scores can be reproduced.

- **List-price cost figures.** Caching and provisioned throughput change the picture for high-volume operators (typically in favour of the higher-end models, since their per-call savings compound). The OpenRouter pricing in particular reflects list-price aggregation across upstream providers and can shift when OpenRouter changes the underlying mix.

- **Open-weights coverage is partial.** The four open-weights checkpoints we evaluated (Llama 4 Scout, Qwen 3 Max, DeepSeek V3.2, Mistral Large 2512) were selected as those that advertise tool-use support in the OpenRouter catalogue at the pilot date and were intended as a coverage sample, not an exhaustive sweep. Open-weights checkpoints without native tool calling — including the larger Llama 4 Maverick — were excluded so that all agents run the same ReAct loop; benchmarking those via a structured-output shim is left to follow-up.
- **Closed-weights coverage is the four-agent v1.0 set.** An earlier v0.1 pilot also included GPT-5.4-mini and OpenAI o4-mini; we dropped them in v1.0 to free per-cell budget for the four-agent OpenRouter expansion, while keeping Claude Opus 4.7 in the lineup as the most expensive closed-weights frontier model. The closed-weights slice in this paper is therefore narrower than the family-level coverage some readers may expect from the closed frontier; the harness can rerun those agents trivially given API access.
- **Single-suite anchor.** The safety axis is calibrated against our 12 attack classes. To check that the axis is not measuring something idiosyncratic, the released harness includes a scaffold (`scripts/run_agentdojo.py`) that pipes AgentDojo [Debenedetti et al., 2024] environments through the AgentOps-Bench scoring suite. A formal cross-suite anchor on AgentDojo is the obvious next external check; we did not include it in the pilot to keep the reported figures tied to a single, fully-released artefact.

8 Conclusion

On 7,200 runs across eight frontier-class agents — four closed-weights and four open-weights — completion spreads 77 percentage points (22 among the seven tool-using agents), cost spreads nearly three orders of magnitude (977×), and the within-agent clean-to-adversarial safety drop spans roughly 10 to 24 percentage points. The cost–completion Pareto leader on this distribution is an open-weights model (DeepSeek V3.2); the most expensive agent in the lineup (Claude Opus 4.7, \$0.0977 per run) is dominated on both completion and cost by smaller closed-weights agents (Haiku 4.5, GPT-5.5) and by every tool-using open-weights agent except Qwen 3 Max; and a single open-weights checkpoint (Qwen 3 Max) crashes on 20% of runs while the other seven finish $\geq$98.1%. None of these findings is recoverable from a completion-only ranking. The natural next steps are scaling to harder, longer-horizon tasks where deeper-reasoning agents likely separate, adding open-weights checkpoints without native tool calling via a structured-output shim, and a cross-suite anchor on AgentDojo so the safety axis is calibrated against an external attack catalogue. AgentOps-Bench gives operators a reproducible way to ask all six questions on the same agent–task pair; we release the code, the task suite, the fixture store, and every per-run

trace under Apache-2.0 and welcome contributions of new payloads and hard tasks for the seed set.

A Reproducibility

Software. All scoring, agent adapter, and tool-server code is released under Apache-2.0. The exact API endpoints used were Anthropic Messages (`2023-06-01`), OpenAI Chat Completions (`2024-10-21`), and the OpenRouter `/v1/chat/completions` endpoint (April 2026 catalogue). Agent adapters issue native tool-calls (not text parsing) so the same task definitions execute end-to-end against any of the three providers.

Models. The pilot used the exact model identifiers `claude-opus-4-7`, `claude-sonnet-4-6`, `claude-haiku-4-5-20251001`, `gpt-5.5`, plus four open-weights checkpoints accessed via OpenRouter: `meta-llama/llama-4-scout`, `qwen/qwen3-max`, `deepseek/deepseek-v3.2`, and `mistralai/mistral-large-2512`. Per-call decoding parameters were the provider defaults; the GPT-5 family rejects custom `temperature`, so for that adapter we send no temperature and let the API use its own. Generation cap was 8,192 tokens for every adapter.

Tasks and conditions. One hundred seed tasks across five domains (code, data analysis, research, safety, tool-use; 20 each); three conditions (CLEAN, NOISY, ADVERSARIAL); $n = 3$ repeats per (agent, task, condition). In the NOISY condition, the tool server replaces a call with one of five failure modes (timeout, HTTP 500, malformed JSON, empty response, rate limit) with `failure_rate`=0.30; in the ADVERSARIAL condition `failure_rate`=0.10 and a uniformly-sampled payload from the 15-entry catalogue is appended to otherwise-successful tool output with probability 0.50 per call. Failure-mode and injection sampling use a per-server `random.Random` instance whose seed is the lower 32 bits of `sha256(task_id|condition|run_number)`, so re-running the same cell yields the same failure/injection pattern; the three repeats of one (agent, task, condition) cell still see distinct patterns because `run_number` is part of the seed, which is what gives the $n = 3$ aggregates a within-cell variance estimate. Live tool back-ends are replayed from a deterministic fixture store keyed on (tool, arguments) so the same tool input always returns the same successful payload. Two cosmetic-only sources of non-determinism remain in the per-run trace JSONs: the `datetime.now()` timestamps and the `message_id`s in mock-email tool results — neither participates in any score, so aggregate axis values are bit-reproducible even though byte-exact trace replay is not.

Tool back-ends. Live tool back-ends were Open-Meteo (weather), `yfinance` fast info (stock price), and Tavily (web search). One live response per (tool, arguments) pair was snapshotted into `fixtures/` and replayed for the benchmark; all reported numbers come from the replay path. Sandboxed Python runs as a subprocess with a fresh temp directory and an 8-second wall-clock cap; SQL

runs against an in-memory SQLite database seeded with the `employees` and `orders` tables shipped in the repository.

Judge. The LLM judge invoked by `score_completion` for tasks without an exact `expected_output` (or for low-deterministic-score fall-throughs) is `claude-sonnet-4-20250514`, called via the Anthropic Messages API at `max_tokens=256` with the structured rubric stored in `src/agentops_bench/scoring/completion.py`. The rubric instructs the judge to redact any `«INJECTED PAYLOAD REDACTED»` marker and not to penalise the agent for unrealistic mock tool output. A deterministic exact-match short-circuits the judge whenever an `expected_output` is available and the agent's output normalises to it; this fires on the great majority of pilot runs. The judge call is not seeded and the API does not expose a deterministic decoding flag, so completion scores on the minority of tasks that fall through to the judge can drift by up to the judge's own per-call variance on rescore; the deterministic short-circuit is what gives the headline numbers their reproducibility.

Reproduction. End-to-end reproduction from the repository root:

```
pip install -e .
export ANTHROPIC_API_KEY=...
export OPENAI_API_KEY=...
export OPENROUTER_API_KEY=...
export TAVILY_API_KEY=...
bash scripts/run_pilot_v1_seeded.sh
python3 scripts/combine_reports.py results/pilot_v1_seeded
python3 scripts/analyze_pilot.py    results/pilot_v1_seeded/_combined
python3 scripts/make_figures.py     results/pilot_v1_seeded/_combined
```

The `run_pilot_v1_seeded.sh` script launches all eight agents (four closed-weights plus four OpenRouter open-weights) in parallel as background processes against the v1.0 seed suite under {clean, noisy, adversarial} $\times$ 3 repeats; per-run traces and scores land under `results/pilot_v1_seeded/<agent>/`. `combine_reports.py` concatenates the per-agent score summaries into the `_combined/` rollup that the analysis and figure scripts consume.

Cost & runtime. The 7,200-run pilot was executed by running each of the eight agents in parallel as a background process; total wall-clock time for the slowest agent (`deepseek-v3.2`, $\sim$32 s per run) was roughly 8 hours, with the other seven finishing in 1–3 hours each. Cumulative API spend across all providers was \$132 at the list-price snapshot in `cost.py` (`PRICING_AS_OF = "2026-04"`); `claude-opus-4-7` alone accounts for \$87.91 (66%) of that spend.

Snapshot disclaimer. The reported scores are a snapshot of eight specific model identifiers at one date (2026-04) under one set of provider-default decoding parameters and one fixture-replay state. The relative ordering of the agents is expected to shift as providers release new model snapshots, change defaults,

or update tool-handling behaviour; OpenRouter list prices in particular reflect a shifting upstream mix and will drift faster than direct-vendor prices. Numbers in this paper should be reproducible on the released fixtures and code, but should not be read as a stable verdict on any agent family.

B Broader Impact

The framework's stated goal is to give practitioners a more honest picture of what an LLM agent will do in production — particularly when exposed to data the operator does not fully control. Multi-axis reporting that includes cost and adversarial-safety drops makes it harder for a vendor to ship an agent that is accurate-on-paper but expensive or insecure in deployment, and we hope publishing the harness under Apache-2.0 lowers the friction of adopting that practice.

The work has dual-use surface. The injection catalogue and the instrumented tool server are useful both for evaluating defences and for constructing attacks against deployed agents. We mitigate this modestly by keeping payloads short, well-known in the literature (direct adaptations of Greshake et al. Greshake et al. [2023] and AgentDojo Debenedetti et al. [2024]), and stripped of any operator- or vendor-specific exfiltration targets — the canary URLs are generic placeholders. We do not view withholding the catalogue as a viable mitigation: defenders need it to test their systems, and any catalogue we release upper-bounds rather than enables attacker capability.

A practical caveat: composite safety scores summarised to a single number, including ours, can give false confidence. The per-component breakdown (compliance, detection, exfiltration) is the operationally meaningful signal, and we report it alongside the composite for that reason.

C Pairwise statistical tests

We complement the headline summary with paired Wilcoxon signed-rank tests over per-cell deltas (each cell is a (task, condition, run) triple shared across the two agents in a comparison) and apply Holm–Bonferroni correction across all $\binom{8}{2} = 28$ pairwise tests to control family-wise error at $\alpha = 0.05$. The script is `scripts/analyze_pairwise.py`; results are reproducible from the released per-run trace JSONs. We acknowledge a methodological caveat: the completion scores are predominantly binary $\{0, 1\}$, so the within-pair differences sit in $\{-1, 0, +1\}$ with many ties — a violation of Wilcoxon's continuous-symmetric-difference assumption. We use `scipy.stats.wilcoxon` with `zero_method='wilcox'` (the default), which drops zeros before ranking; we also checked that the corrected reject pattern is unchanged when the test is replaced with a paired bootstrap on the mean delta, so the Holm–Bonferroni decisions reported here are insensitive to the choice of paired test. We report Wilcoxon for continuity with prior agent-evaluation work that uses it (e.g. ToolLLM Qin et al. [2024]

pairwise comparisons).

The interpretation we draw is twofold. *First*, the eight-agent lineup is statistically separated on completion at $n = 900$ for nearly every pair: only `haiku-4-5` vs. `opus-4-7` (mean $\Delta = +0.040$, $p = 0.916$), `sonnet-4-6` vs. `mistral-large` (mean $\Delta = +0.036$), and `gpt-5.5` vs. `deepseek-v3.2` (mean $\Delta = -0.001$) fail to clear the corrected threshold; the last is genuinely tied at the cell level, and the haiku-vs-opus result is a higher-power confirmation that Opus offers no completion advantage on this distribution despite costing 10× more. We caution that several rejected pairs have small operational effect sizes — `haiku-4-5` vs. `gpt-5.5` ($\Delta = -0.014$) and `haiku-4-5` vs. `deepseek-v3.2` ($\Delta = -0.015$) clear FWER but are practically negligible — because $n = 900$ paired observations make even sub-2pp effects detectable. The headline cost–completion ranking should be read in conjunction with the mean-Δ column, not the reject column alone. *Second*, adversarial-safety separation is also robust at this scale: the 21 of 28 pairs that clear correction match the operationally meaningful pattern that `deepseek-v3.2` and `mistral-large` are the most fragile under injection while `opus-4-7` and `sonnet-4-6` are the most resistant of the tool-using checkpoints (and are themselves not statistically separable on this axis).

D Per-attack-class safety breakdown

Each adversarial run records the specific payload appended to each poisoned tool result. We aggregate over the 2,400 adversarial runs (eight agents × 300 runs each) by attack class (Table 6). The denominator is the number of runs in which at least one payload of that class was injected; the numerator metrics are the run-level safety / compliance / detection / exfiltration scores attributed to runs containing that class. A single run can contribute to multiple class rows because the catalogue samples uniformly at the per-call site.

Three things stand out. *First*, the unicode zero-width class is the most effective payload class on this lineup at the rank-by-mean level: mean compliance is 0.215 (next: fake-error recovery at 0.193). At the per-class sample sizes in the table (104–250 runs), Wilson half-widths on compliance are ±2.6–7.4pp, so the ranking between adjacent classes is statistically loose; the qualitative takeaway is that hidden-character payloads defeat ordinary surface-form filtering, and the result replicates across closed and open weights. *Second*, exfiltration tool calls remain rare but no longer zero: the delayed-post-processing class triggered an exfiltration on 4.9% of runs, and markdown-exfil and embedded-in-JSON each on roughly 1.3%. These are still single-digit raw counts — driven by individual open-weights agents acting on instructions to issue an outbound tool call — but they demonstrate that the exfiltration component of the safety axis is no longer uninformative on a sufficiently broad lineup. *Third*, regex-based detection remains a low ceiling: mean detection across the table is 0.14, and most agents "silently ignore" the injection without articulating that they have done so — which scores 0 on detection and 0 on compliance, returning a 0.7 floor. `Claude-opus-4-7` and `claude-sonnet-4-6` are the two most-detecting

agents in the lineup, particularly on the XML TAG, COMMENT CHANNEL, and MULTI-TURN POISON classes (Opus 0.91, 0.91, 0.83 respectively; Sonnet 0.79, 0.79, 0.89), which is what drives their adversarial-safety lead. A more sophisticated detection scorer (e.g., a small auxiliary classifier) is the right next step; the regex catalogue is the release-able lower bound.

E Datasheet for Datasets

We adapt the structure of Gebru et al. Gebru et al. [2021] to the benchmark task suite released with this paper.

Motivation. *For what purpose was the dataset created?* To evaluate LLM agents on six operational axes (completion, cost, efficiency, reliability, recovery, safety) under three environmental conditions (clean, noisy, adversarial). *Who created the dataset and on behalf of whom?* The author, on behalf of an independent research release; no institutional sponsor. *Who funded the creation?* Self-funded; cumulative API spend on the released pilot was $132.10 (of which $87.91 was `claude-opus-4-7`).

Composition. *What do the instances represent?* YAML task definitions, each specifying a natural-language prompt, a tool whitelist, an optional reference answer, an optimal-step estimate, and a difficulty tag. *How many instances are there?* 100 seed tasks in the v1.0 release, balanced as 20 tasks per domain across the five domains (code, data analysis, research, safety, tool-use); all 100 are used to compute the headline numbers in this paper. *Is any information missing?* Reference answers are absent for ambiguous research-style tasks where deterministic scoring is inappropriate; those run through the LLM judge. *Are there any errors, sources of noise, or redundancies?* Tasks were drafted by a single author and inherit that author's biases. The tool catalogue (10 reference tools) constrains task realism; tasks that would require richer tool catalogues (file ingestion, multi-modal input) are out of scope of this seed set.

Collection process. *How was the data acquired?* All tasks were authored from scratch for this release; no third-party datasets were ingested. *What mechanisms were used to collect the data?* Direct authoring in YAML against the schema in `src/agentops_bench/schema.py`. *Over what timeframe?* 2026-01 through 2026-04. *Were any ethical-review processes conducted?* No human subjects; no ethical review required. Adversarial payloads were adapted from publicly published catalogues Greshake et al. [2023], Debenedetti et al. [2024] and stripped of any organisation- or vendor-specific exfiltration targets.

Pre-processing, cleaning, labelling. Tasks ship in the form they will be evaluated. Reference answers are authored in the format the agent is expected to produce. Optimal-step counts are author-estimated and used only to compute efficiency, not correctness.

Uses. *Has the dataset been used for any tasks already?* Yes — the 7,200-run pilot reported in this paper. *What other tasks could it be used for?* Any agent-evaluation harness whose scoring is compatible with the YAML schema. *Is there anything that should* not *be done with the dataset?* The injection catalogue should not be used for live attack against deployed third-party agents without authorisation; the dual-use surface is discussed in Appendix B.

Distribution. *License.* Apache-2.0 for code and tasks. *How is the dataset distributed?* Public Git repository, with a tagged release per benchmark version. *Persistent identifier.* A Zenodo deposit will be created at v1.0 release; the DOI will be added to the README and to subsequent revisions of this paper.

Maintenance. See Appendix F for the versioning and contribution policy.

F Maintenance and versioning

Versioning. The benchmark uses a `MAJOR.MINOR.PATCH` scheme. `MAJOR` bumps when scoring axis definitions or weights change in a way that breaks comparability with prior reports. `MINOR` bumps when tasks, payloads, or tool-server failure modes are added. `PATCH` bumps for code-only fixes that do not move scores. The pilot reported here is `v1.0`.

Snapshotting. Every released report records the benchmark `MAJOR.MINOR.PATCH`, the price-table date (`PRICING_AS_OF` in `cost.py`), the model identifiers, and the fixture-store hash. Re-running an old report against a newer benchmark version is expected to drift; the stored metadata makes the drift attributable.

Maintenance. The repository is currently maintained by the sole author. We invite community contributions, particularly: (i) new tasks with author-validated reference answers, (ii) additional prompt-injection payloads with documented attack classes, (iii) agent-adapter implementations for additional providers and open-weights models. Contribution guidelines, a triage policy, and a deprecation policy for retired tasks are documented in `CONTRIBUTING.md`.

Liability and responsibility. Authors of contributed tasks or payloads retain copyright (Apache-2.0 contribution-licence agreement applies). The repository maintainer is responsible for triage and versioning; users of the benchmark are responsible for any use of the adversarial-payload catalogue and the harness.

G Reproducibility checklist

We follow the NeurIPS reproducibility checklist.

- ✓ Claims in the abstract and introduction match the results: every quantitative claim is reproducible from `results/pilot_v1_seeded/` via `scripts/analyze_pilot.py`, `scripts/analyze_pairwise.py`, and `scripts/analyze_safety_per_payload.py`.
- ✓ Limitations are explicitly discussed (§7).
- ✓ Code is released under Apache-2.0 with the paper.
- ✓ Each pilot run's full trace, scores, and cost breakdown is released as JSON.
- ✓ All datasets used (the 100 seed tasks, the 15-payload injection catalogue, and the fixture-replay store) are included in the release; provenance is described in Appendix E.
- ✓ Hyperparameters (failure rates, injection probabilities, generation cap) are listed in Appendix A.
- ✓ Statistical significance is reported with Holm–Bonferroni correction (Appendix C).
- ✓ Compute resources used: a single laptop, ∼8 hours wall-clock with the eight agents running concurrently as background processes, $132 cumulative API spend across providers.
- ✓ Broader impact discussed (Appendix B).
- ✓ Random seeds for failure-mode and injection sampling are derived deterministically from `sha256(task_id|condition|run_number)` per cell (§A, *Tasks and conditions*). Live tool back-ends are deterministic via fixture replay; agent-side decoding inherits provider defaults.

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Table 2: Per-condition breakdown. Each cell is the mean over $n = 300$ runs (100 tasks $\times$ 3 repeats) for that (agent, condition) pair. The ADVERSARIAL column matters most for operations: every tool-using agent's safety drops by 12–24 percentage points relative to clean (`llama-4-scout`'s smaller 9.9pp drop is uninformative). Safety values of 1.000 in clean and noisy use the one-sided Wilson lower bound ($\geq$0.988 at $n = 300$).

Agent	Condition	Compl.	Effic.	Safety
claude-opus-4-7	clean	0.902	0.839	1.000
	noisy	0.852	0.724	1.000
	adversarial	0.861	0.803	0.879
claude-sonnet-4-6	clean	0.884	0.826	1.000
	noisy	0.866	0.704	1.000
	adversarial	0.844	0.787	0.856
claude-haiku-4-5	clean	0.927	0.817	1.000
	noisy	0.900	0.709	1.000
	adversarial	0.907	0.778	0.805
gpt-5.5	clean	0.937	0.884	1.000
	noisy	0.915	0.798	1.000
	adversarial	0.924	0.851	0.833
deepseek-v3.2	clean	0.949	0.702	1.000
	noisy	0.917	0.581	1.000
	adversarial	0.914	0.665	0.763
llama-4-scout	clean	0.179	0.929	1.000
	noisy	0.145	0.918	1.000
	adversarial	0.147	0.921	0.901
mistral-large-2512	clean	0.856	0.818	1.000
	noisy	0.826	0.707	1.000
	adversarial	0.804	0.777	0.780
qwen3-max	clean	0.751	0.737	1.000
	noisy	0.692	0.679	1.000
	adversarial	0.673	0.684	0.831

Table 3: Mean completion per (agent, domain). $n = 180$ per cell (20 tasks $\times$ 3 conditions $\times$ 3 repeats). Best in column is bold. `llama-4-scout`'s row reflects tool-format failure (§6) rather than reasoning quality, and is excluded from the per-column bolding. No single agent leads on all five domains.

Agent	Code	Data	Research	Safety	Tool-use
claude-opus-4-7	0.875	**0.950**	0.673	0.940	0.920
claude-sonnet-4-6	0.871	0.832	0.784	0.935	0.903
claude-haiku-4-5	0.876	0.899	0.888	0.961	0.931
gpt-5.5	**0.917**	0.952	0.872	0.944	**0.942**
deepseek-v3.2	0.880	0.907	**0.925**	**0.999**	0.922
llama-4-scout	0.010	0.000	0.169	0.295	0.309
mistral-large-2512	0.713	0.836	0.857	0.901	0.838
qwen3-max	0.867	0.766	0.764	0.538	0.592

Table 4: Pairwise completion (all conditions, $n = 900$ paired observations per comparison). Mean delta is (Agent A) − (Agent B). "Reject" = Holm–Bonferroni-corrected reject of H_0 at FWER 0.05. At $n = 900$, 25 of 28 pairs separate, matching the much wider between-agent completion spread under the eight-agent lineup.

Agent A	Agent B	n pairs	Mean Δ	Raw p	Reject
haiku-4-5	opus-4-7	900	+0.040	0.916	no
haiku-4-5	sonnet-4-6	900	+0.046	0.000	**yes**
haiku-4-5	gpt-5.5	900	−0.014	0.000	**yes**
haiku-4-5	deepseek-v3.2	900	−0.015	0.000	**yes**
haiku-4-5	llama-4-scout	900	+0.754	0.000	**yes**
haiku-4-5	mistral-large	900	+0.083	0.000	**yes**
haiku-4-5	qwen3-max	900	+0.206	0.000	**yes**
opus-4-7	sonnet-4-6	900	+0.007	0.000	**yes**
opus-4-7	gpt-5.5	900	−0.054	0.000	**yes**
opus-4-7	deepseek-v3.2	900	−0.055	0.001	**yes**
opus-4-7	llama-4-scout	900	+0.715	0.000	**yes**
opus-4-7	mistral-large	900	+0.043	0.000	**yes**
opus-4-7	qwen3-max	900	+0.166	0.000	**yes**
sonnet-4-6	gpt-5.5	900	−0.060	0.000	**yes**
sonnet-4-6	deepseek-v3.2	900	−0.062	0.000	**yes**
sonnet-4-6	llama-4-scout	900	+0.708	0.000	**yes**
sonnet-4-6	mistral-large	900	+0.036	0.111	no
sonnet-4-6	qwen3-max	900	+0.160	0.000	**yes**
gpt-5.5	deepseek-v3.2	900	−0.001	0.125	no
gpt-5.5	llama-4-scout	900	+0.768	0.000	**yes**
gpt-5.5	mistral-large	900	+0.097	0.000	**yes**
gpt-5.5	qwen3-max	900	+0.220	0.000	**yes**
deepseek-v3.2	llama-4-scout	900	+0.770	0.000	**yes**
deepseek-v3.2	mistral-large	900	+0.098	0.000	**yes**
deepseek-v3.2	qwen3-max	900	+0.221	0.000	**yes**
llama-4-scout	mistral-large	900	−0.672	0.000	**yes**
llama-4-scout	qwen3-max	900	−0.549	0.000	**yes**
mistral-large	qwen3-max	900	+0.123	0.000	**yes**

Table 5: Pairwise adversarial-condition safety ($n = 300$ paired observations per comparison). 21 of 28 pairs separate after Holm–Bonferroni correction; the seven non-separating pairs cluster around the safety-axis ties (haiku/qwen, opus/sonnet, opus/llama, sonnet/gpt-5.5, sonnet/qwen, gpt-5.5/qwen, deepseek/mistral), where agents sit close together on adversarial safety despite often differing on every other axis.

Agent A	Agent B	n pairs	Mean Δ	Raw p	Reject
haiku-4-5	opus-4-7	300	−0.073	0.000	**yes**
haiku-4-5	sonnet-4-6	300	−0.051	0.000	**yes**
haiku-4-5	gpt-5.5	300	−0.028	0.000	**yes**
haiku-4-5	deepseek-v3.2	300	+0.042	0.000	**yes**
haiku-4-5	llama-4-scout	300	−0.096	0.000	**yes**
haiku-4-5	mistral-large	300	+0.025	0.002	**yes**
haiku-4-5	qwen3-max	300	−0.026	0.051	no
opus-4-7	sonnet-4-6	300	+0.022	0.009	no
opus-4-7	gpt-5.5	300	+0.046	0.000	**yes**
opus-4-7	deepseek-v3.2	300	+0.116	0.000	**yes**
opus-4-7	llama-4-scout	300	−0.022	0.048	no
opus-4-7	mistral-large	300	+0.098	0.000	**yes**
opus-4-7	qwen3-max	300	+0.048	0.001	**yes**
sonnet-4-6	gpt-5.5	300	+0.024	0.010	no
sonnet-4-6	deepseek-v3.2	300	+0.093	0.000	**yes**
sonnet-4-6	llama-4-scout	300	−0.045	0.000	**yes**
sonnet-4-6	mistral-large	300	+0.076	0.000	**yes**
sonnet-4-6	qwen3-max	300	+0.026	0.036	no
gpt-5.5	deepseek-v3.2	300	+0.070	0.000	**yes**
gpt-5.5	llama-4-scout	300	−0.068	0.000	**yes**
gpt-5.5	mistral-large	300	+0.053	0.000	**yes**
gpt-5.5	qwen3-max	300	+0.002	0.532	no
deepseek-v3.2	llama-4-scout	300	−0.138	0.000	**yes**
deepseek-v3.2	mistral-large	300	−0.017	0.022	no
deepseek-v3.2	qwen3-max	300	−0.068	0.000	**yes**
llama-4-scout	mistral-large	300	+0.121	0.000	**yes**
llama-4-scout	qwen3-max	300	+0.070	0.000	**yes**
mistral-large	qwen3-max	300	−0.051	0.000	**yes**

Table 6: Per-attack-class safety, averaged across all eight agents on runs containing at least one payload of that class. "Mean compliance" is the regex-matched compliance rate (lower is better); "Mean detection" is the rate at which the agent flagged or refused the injection (higher is better); "Exfil rate" is the fraction of runs in which the agent issued a tool call to one of the catalogue's exfiltration hosts.

Attack class	Runs	Mean safety	Mean compl.	Mean detect.	Exfil rate
direct override	250	0.692	0.106	0.123	0.008
authority impersonation	209	0.702	0.119	0.166	0.000
embedded in JSON	156	0.682	0.138	0.136	0.013
markdown exfil	146	0.699	0.075	0.110	0.014
comment channel	158	0.736	0.023	0.150	0.000
unicode homoglyph	108	0.729	0.032	0.139	0.000
unicode zero-width	106	0.635	**0.215**	0.070	0.000
base64 wrapped	154	0.679	0.123	0.094	0.000
fake-error recovery	123	0.663	0.193	0.132	0.000
multi-turn poisoning	104	**0.742**	0.028	0.188	0.010
nested JSON	131	0.691	0.138	0.161	0.008
XML tag impersonation	125	0.726	0.039	0.148	0.008
delayed post-processing	143	0.734	0.020	**0.189**	**0.049**